ChandrikaM

Lab 06-09

```
  GNU nano 7.2                                                    OS.sh
#!/bin/bash

echo "Lab 6: Directory Management"

while true
do

   echo ""
   echo "1. Create Directory"
   echo "2. List Directories"
   echo "3. Delete Directory"
   echo #4. Exit"

   echo "Enter your choice:"
   read choice

   case $choice in

1)    echo "Enter directory name to create:"

      read dirname

      mkdir "$dirname"

      if [ $? -eq]; then
      echo "Directory '$dirname' created successfully."

      else
      echo "Error creating directory."
   fi
   ;;


2)

      echo "Directories in current location:"
                                                                [ Read 59 lines ]
```

```
2)

      echo "Directories in current location:"

      ls -d */
   ;;
3)
     echo "Enter Directory name to delete:"
     read dirname
     rmdir "$dirname"
     if [ $? -eq 0 ]; then
         echo "Directory '$dirname' deleted successfully."
     else
         echo "Error deleting directory (may not be empty)."
     fi
     ;;
4)
     echo "Exiting program..."
     break
     ;;
*)
     echo "Invalid choice. Try again!"
     ;;
  esac
done
```

```
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ nano OS.sh
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ ./OS.sh
Lab 6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory

Enter your choice:
2
Directories in current location:
ls: cannot access '*/': No such file or directory

1. Create Directory
2. List Directories
3. Delete Directory

Enter your choice:
1
Enter directory name to create:
CHandu
./OS.sh: line 25: [: missing `]'
Error creating directory.

1. Create Directory
2. List Directories
3. Delete Directory

Enter your choice:
3
Enter Directory name to delete:
Chandu
Directory 'Chandu' deleted successfully.

1. Create Directory
2. List Directories
3. Delete Directory

Enter your choice:
4
Exiting program...
```

```
  GNU nano 7.2
echo "Lab 7:Memory Usage Monitor"

echo ""
echo "Current Memory Status (using free command):"
free -h

echo ""
echo "Detailed Memory Info (using top command):"
top -b -n 1 | grep "MiB Mem"

echo ""
echo "Available Memory:"
free -h |grep "Mem" \ awk '{print "Available : " $7}'
```

```
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ nano OS7.sh
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ ./OS7.sh
Lab 7:Memory Usage Monitor

Current Memory Status (using free command):
              total        used        free      shared  buff/cache   available
Mem:          3.8Gi       412Mi       3.4Gi       3.4Mi       103Mi       3.4Gi
Swap:         1.0Gi          0B       1.0Gi

Detailed Memory Info (using top command):
MiB Mem :   3853.2 total,   3453.8 free,    415.4 used,    104.0 buff/cache

Available Memory:
grep:  awk: No such file or directory
grep: {print "Available : " $7}: No such file or directory
```

```
  GNU nano 7.2                                                    OS8.sh
echo "Lab 8: Environment Variable Demo "

# Setting Variable
name="Chandrika"
course="Operating Systems"

echo ""
echo "Local Variables:"
echo "Name: $name"
echo "Course: $course"

# Exporting variables
export name
export course


echo""
echo "Using printenv command:"
printenv name
printenv course

echo ""
echo "All Environment Variable (using env):"
env
```

```
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ nano OS8.sh
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ ./OS8.sh
Lab 8: Environment Variable Demo

Local Variables:
Name: Chandrika
Course: Operating Systems

Using printenv command:
Chandrika
Operating Systems

All Environment Variable (using env):
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu-24.04
WT_SESSION=d4461e36-5626-48c4-a5ee-c8e7be892688
NAME=CHANDUREDDY
PWD=/mnt/c/Users/Chandu/OneDrive/Desktop/OS
LOGNAME=chandu
HOME=/home/chandu
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/292_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lz
ma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*
.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*
.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:
*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=
00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*~=00;90:*#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00
;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=chandu
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Intel/I
ntel(R) Management Engine Components/iCLS/:/mnt/c/Program Files/Intel/Intel(R) Management Engine Components/iCLS/:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/
c/Program Files (x86)/Intel/Intel(R) Management Engine Components/DAL:/mnt/c/Program Files/Intel/Intel(R) Management Engine Components/DAL:/mnt/c/Program Files/Intel/WiFi/bin/:/mnt/c/Program Files/Common Files/Intel/WirelessCommon/:/mnt/c/Program Files/Git/cmd:/mnt/c/
Users/Chandu/AppData/Local/Programs/Python/Python313/Scripts/:/mnt/c/Users/Chandu/AppData/Local/Programs/Python/Python313/:/mnt/c/Users/Chandu/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Chandu/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Chandu/AppD
ata/Local/Programs/Windsurf/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=Operating Systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={62c98c74-bffa-588c-b2cd-1d74fb98c54b}
OLDPWD=/mnt/c/Users/Chandu/OneDrive/Desktop
name=Chandrika
_=/usr/bin/env
```

```
  GNU nano 7.2                                                    OS9.sh
echo "Lab 09 : I/0 Redirection Demo"

#output redirection (Overwrite)
echo "Hello, This is OS Lab" >file1.txt
echo "Content written to file1.txt"

#Append output
echo "Appending new line">> file1.txt
echo "Content append to file1.txt"

#Input redirection
echo ""
echo "Trying to read non-existing file:"
cat file_not_exist.txt 2> error.txt
echo "error message stored in error.txt"

#display error file
echo ""
echo"Error file content:"
cat error.txt
```

```
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ nano OS9.sh
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$ ./OS9.sh
Lab 09 : I/0 Redirection Demo
Content written to file1.txt
Content append to file1.txt

Trying to read non-existing file:
error message stored in error.txt

./OS9.sh: line 19: echoError file content:: command not found
cat: file_not_exist.txt: No such file or directory
chandu@CHANDUREDDY:/mnt/c/Users/Chandu/OneDrive/Desktop/OS$
```